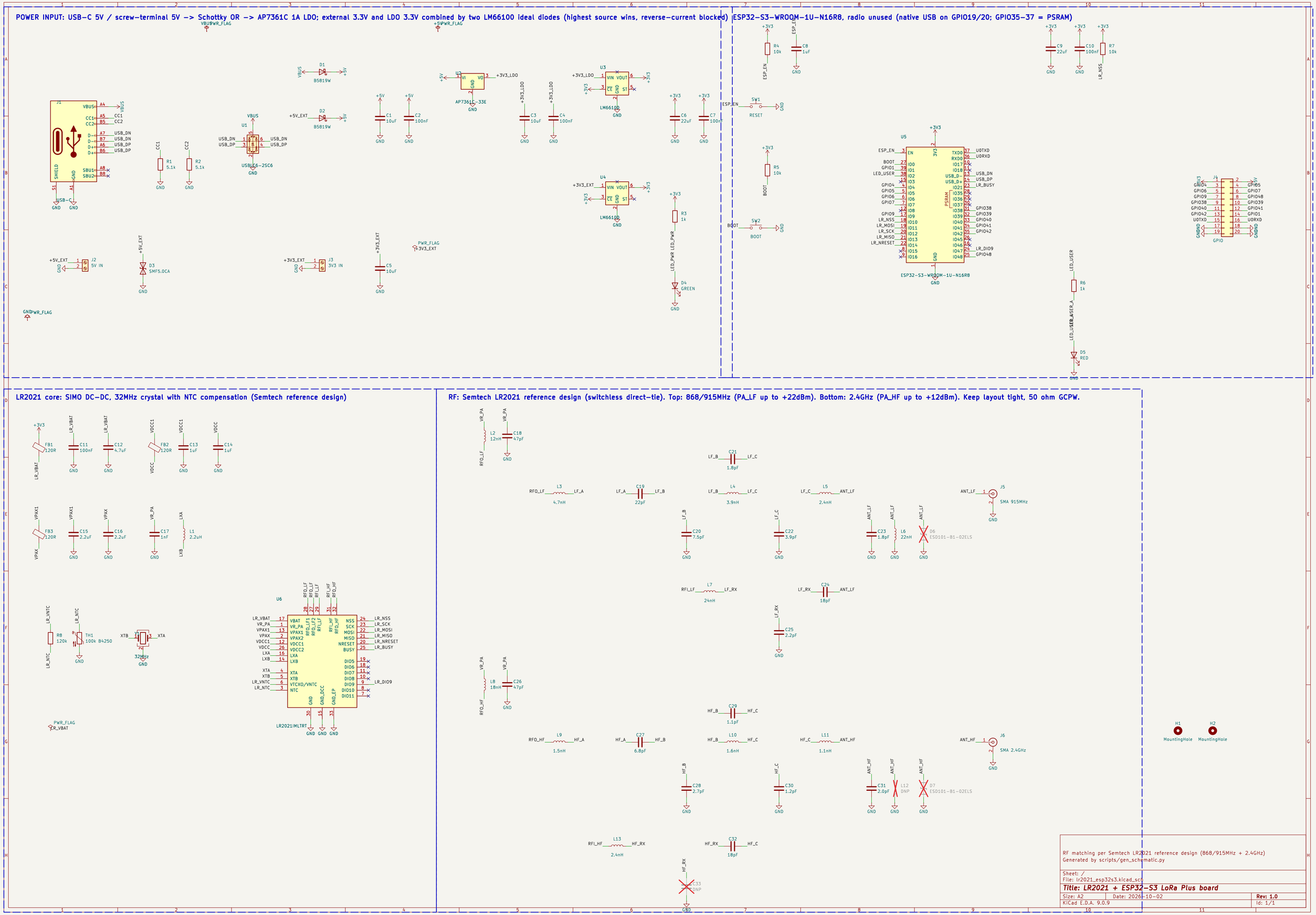
POWER INPUT: USB-C 5V / screw-terminal 5V -> Schottky OR -> AP7361C 1A LDO; external 3.3V and LDO 3.3V combined by two LM66100 ideal diodes (highest source wins, reverse-current blocked)

ESP32-S3-WROOM-1U-N16R8, radio unused (native USB on GPIO19/20; GPIO35-37 = PSRAM)




LR2021 core: SIMO DC-DC, 32MHz crystal with NTC compensation (Semtech reference design)




RF: Semtech LR2021 reference design (switchless direct-tie). Top: 868/915MHz (PA_LF up to +22dBm). Bottom: 2.4GHz (PA_HF up to +12dBm). Keep layout tight, 50 ohm GCPW.




RF matching per Semtech LR2021 reference design (868/915MHz + 2.4GHz)
Generated by scripts/gen_schematic.py

Sheet: /
File: lr2021_esp32s3.kicad_sch

**Title: LR2021 + ESP32-S3 LoRa Plus board**

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|---|---|---|
| Size: A2 | Date: 2026-10-02 | Rev: 1.0 |
| KiCad E.D.A. 9.0.9 | | Id: 1/1 |